CS423 – CSC15003 – Software Testing (AI-augmented · 2026)

HOMEWORK – AI-FIRST EDITION (2026 v1.0)

HW01 – QA/QC Jobs · 20 Defects · Test a Physical Product

Exercise ID:	HW01-AI
Student:	Nguyen Duc Quang
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Form:	Individual Assignment
Submission:	Moodle (report) + GitHub repo link (artifacts)
AI Policy:	Open – Mandatory declaration + AI Audit Report
Bloom-AI:	G9.1 (Understand) · G9.3 (Analyse)
Date:	29/09/2026
Self-assessed grade:	**090 / 100**

1. Outcomes (this HW)

- Describe the 2026+ QA/QC job-market landscape (including AI-augmented testing roles).
- Distinguish work that AI can replace / assist / cannot replace in QA/QC.
- Design test cases for a SPECIFIC physical product (not a software product).
- G9.1: ask AI for a QA/QC role mindmap and find ≥ 3 mistakes.
- G9.3: analyse AI-generated test cases and find ≥ 3 missed edge cases.

Note: HW01 is a warm-up — it does **not** use the EShop SUT.

2. Requirement 1 – QA/QC Job Market 2026+ (40 pts)

Collected **10** QA/QC postings within ~60 days of submission. **≥ 5** require AI/LLM/automation-AI skills (Jobs 01–05 marked AI-REQUIRED). Screenshots stored under `evidence/jobs/`.

#	Title / Company	AI?	Salary	Screenshot	AI Impact (short)
01	Staff AI Test Automation Eng. — Gravie	YES	USD 155–206.5k	job01_gravie.png	AI agents are core of the role.
02	Test Architect Agentic AI — Wolters Kluwer	YES	N/D	job02_wolters.png	QA expands to LLM eval / hallucination detection.

03	Sr Automation Test (AI) — Hitachi VN	YES	N/D	job03_hitachi_ai.png	Copilot/Cursor listed in VN JD.
04	QA Intern (AI-Assisted) — BigIn	YES	N/D	job04_bigin.png	Even intern roles require GenAI tools.
05	Senior SDET — Enzo Tech	YES	N/D	job05_enzo.png	AI tools expected in daily testing.
06	Sr Automotive Test Eng. — Hitachi	assist	N/D	job06_hitachi_auto.png	GenAI assists; safety review stays human.
07	Software QA Eng. — Teradyne	no	N/D	job07_teradyne.png	Classic process/API skills dominate.
08	QA Eng. Manual+Auto — Pavago	no	N/D	job08_pavago.png	Hybrid QA; AI not central in JD.
09	QA Eng. Manual+Python — TestUnity	optional	N/D	job09_testunity.png	Agentic testing only good-to-have.
10	Sr QA Manual+Auto — Goods & Services	awareness	N/D	job10_goods_services.png	AI awareness required, not agent building.

2.1 G9.1 QA/QC mindmap

AI draft mindmap (INVALID) and corrected mindmap are in `mindmap/` (PNG + Markdown).

3 mistakes found in AI mindmap:

1. Claimed ISTQB Advanced is required for every Manual QA — incorrect; Foundation / experience is common.
2. Claimed software has no QC, Selenium-only forever, Waterfall-only in 2026 — incorrect.
3. Dropped exploratory testing; reduced AI roles to “prompt copy-paste” — incorrect.

3. Requirement 2 – 20 Software Defects 2022–2026 (20 pts)

Full table: `evidence/defects/defects_2022_2026.md` and `excel/HW01_Defects.csv`. Summary: **20** defects; **6** AI/LLM-related (#1–#6).

#	Year	Title	Type	Severity
1	2022–24	Air Canada chatbot bereavement fare (Moffatt)	AI/LLM	High
2	2023	Samsung ChatGPT source-code paste	AI/LLM	Critical
3	2023	Chevy Watsonville chatbot $1 Tahoe (prompt injection)	AI/LLM	High

4	2023	Google Bard JWST demo error	AI/LLM	Medium
5	2023	Bing Chat “Sydney” unsafe replies	AI/LLM	High
6	2023	ChatGPT Redis chat-history exposure	AI/LLM	Critical
7	2024	CrowdStrike Falcon Channel File 291	Classic	Critical
8	2024	Change Healthcare ransomware (no MFA)	Classic	Critical
9	2022	Southwest Airlines scheduling meltdown	Classic	Critical
10	2022	Optus data breach	Classic	Critical
11	2022	Medibank data breach	Classic	Critical
12	2022	Uber Lapsus$ breach	Classic	High
13	2022	LastPass vault backup theft	Classic	Critical
14	2023	MOVEit CVE-2023-34362	Classic	Critical
15	2023	CitrixBleed CVE-2023-4966	Classic	Critical
16	2023	FAA NOTAM outage	Classic	Critical
17	2023	Microsoft Storm-0558 forged tokens	Classic	Critical
18	2024	xz Utils backdoor CVE-2024-3094	Classic	Critical
19	2024	Snowflake credential stuffing	Classic	Critical
20	2022	Atlassian cloud major outage	Classic	High

3.1 AI hallucination when explaining a defect (mandatory)

Defect chosen: Air Canada chatbot (*Moffatt v. Air Canada*). Same NVD-style trap prompt:

- **ChatGPT:** correctly refused to invent CVE/CVSS (VALID / careful).
- **Gemini:** invented `CVE-2022-99999`, CVSS 5.3, product `Air_Canada_Conversational_Web_Chatbot_v1.0` → **INVALID** (Audit Artifact A3e).

4. Requirement 3 – Physical product test cases (25 pts in AI-first rubric)

Field	Value
Device	SHARP humidifying air purifier
Model	KC-A70-W
Year	2012
Serial (masked)	10****5 (full 1081245)
Photo	evidence/device/device_plus_studentID.jpg (device + student ID same frame)

15 test cases in `excel/HW01_TestCases.csv`. Student edge cases AI missed: **TC13** low water warning, **TC14** power interrupt, **TC15** blocked intake.

Executed + video (≥ 5): TC01, TC02, TC03, TC04, TC07, TC14 — all **Pass**.

#	TC	YouTube (Unlisted)	Verdict
1	TC01 Power ON	youtube.com/shorts/TDc4qjqxIPg	Pass
2	TC02 Power OFF	youtube.com/shorts/6TNnUrJ88wM	Pass
3	TC03 Humidity Display	youtube.com/shorts/TrcpG13t7ZY	Pass
4	TC04 Humidify Water Lamp	youtube.com/shorts/8eSd7PK7YQE	Pass
5	TC07 Airflow Outlet	youtube.com/shorts/1j9dfo1A-hU	Pass
6	TC14 Power Interrupt	youtube.com/shorts/CzGJKSu_WKQ	Pass

5. AI Collaboration Protocol

5.1 CLO mapping

CLO	Activity in this HW
G9.1	AI mindmap of QA/QC roles; student found 3 mistakes and corrected mindmap.
G9.3	AI-generated TCs analysed; student added ≥ 3 edge cases (TC13–TC15).

5.2 Tools used

Tool	Purpose
Cursor (Agent)	Scaffold, mindmap, TC drafts, report packaging
ChatGPT	Defect Q&A; CVE/NVD trap prompts
Gemini	Same NVD trap → hallucination sample

5.3 AI Audit Report (summary) — full 5-section entries in Appendix / AI-02

Artifact	Verdict
A1 Mindmap	INVALID
A2 Physical TCs	INCOMPLETE
A3 Air Canada explanation	INCOMPLETE (weak sources)
A3b–A3d ChatGPT traps	VALID (refused to invent CVE)
A3e Gemini NVD table	INVALID (hallucinated CVE/CVSS)

Verdict	Count	%
VALID	3	43%
INVALID	2	29%
INCOMPLETE	2	29%
Total	7	100%

When to use AI: drafts, brainstorming, creating hallucination traps.
When NOT to use AI: device+ID photo, voice videos, job login screenshots, signed forms, unverified CVE/facts.

5.4 AI Critique (200–300 words)

In this homework I used Cursor to draft a QA/QC mindmap and physical-product test cases, and I used ChatGPT and Gemini to explain the Air Canada chatbot defect and to stress-test whether models invent CVE or CVSS data. The biggest failure was Gemini: even after saying there is no official NVD entry, it still filled a table with CVE-2022-99999, a fake product version, and a CVSS 5.3 vector. That is a clear hallucination under pressure to "complete every field." ChatGPT on the same prompts often refused and answered "none," which shows tool choice and prompting change the risk.

The mindmap draft was also wrong in a quieter way: it forced ISTQB Advanced for every Manual QA role, claimed software has no QC, and treated AI work as prompt copy-paste while dropping exploratory testing. Those mistakes match incomplete understanding of ISTQB roles and the 2026 job market we collected. For the Sharp KC-A70-W, AI mostly proposed happy-path cases (power, humidity display, indicators) and missed edges such as power interrupt mid-run, empty-tank warning, and blocked intake—so I added TC13–TC15 myself.

The principle I take away is simple: AI is useful for first drafts and for creating traps that expose hallucination, but it is a bad oracle. Every factual cell (CVE, salary, model, severity) needs a primary source or a real device observation. Anti-cheat evidence (ID photo, voice videos, job screenshots) must stay human-made. Collaborating with AI means generate → audit (VALID/INVALID/INCOMPLETE) → fix → log the prompt, not paste and submit.

5.5 Mandatory Disclosure

> Test cases (physical product) and the QA/QC role mindmap were initially generated by **Cursor (Agent)**; defect explanations and hallucination traps used **ChatGPT** and **Gemini**. I reviewed and modified the mindmap (3 corrections), replaced incomplete AI test cases with a final set of 15 and added edge cases **TC13, TC14, TC15**; job collection tables and AI Impact Analysis were drafted with AI assist then edited by me. Job screenshots, device photo with student ID, execution videos with my voice narration, prompt log timestamps of my sessions, and this disclosure were produced entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category below.

6. Self-Assessment Template

No.	Criteria	Max	Self-Assessed
1	Job Market 2026+ (10 jobs × 3 pts + AI Impact)	40	34
2	Software Defects 2022–2026 (20 defects)	20	18
3	Physical-product test design (15 TCs + videos)	25	23
AI-1	[AI-02] AI Audit Report (5-section)	8	8
AI-2	AI Critique 200–300 words + [AI-03] Disclosure	4	4
AI-3	[AI-05] Checklist + anti-cheat artifacts	3	3
	Total	**100**	**90**

Zip filename: `20120359_HW01_AI_090.zip`

7. Appendices (files in submission package)

- Appendix A: `ai/prompt_log.md`
- [AI-02] `ai/templates/AI-02_AI_Audit_Report.md`
- [AI-03] `ai/templates/AI-03_AI_Disclosure_Form.md`
- [AI-05] `ai/templates/AI-05_Privacy_Responsible_Use_Checklist.md`
- Excel/CSV: `excel/HW01_TestCases.csv`, `excel/HW01_Defects.csv`, `excel/HW01_TestSummary.md`
- Mindmap: `mindmap/qa_qc_mindmap_*.png`
- Device evidence: `evidence/device/`
- Job screenshots: `evidence/jobs/`
- Videos: `evidence/videos/youtube_links.md`
- Note: FIT Mantis screenshots — add if course account available (`evidence/mantis/`)

— End of main report —